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When Is 3D Worth It? Task-Conditioned Adequacy of Planar Localisation for Vision-Guided Manipulation

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Abstract— Planar image-to-workspace mappings are attractive in vision-guided manipulation because they are simple, inexpensive, and easy to calibrate, yet their geometric assumptions are violated when targets depart from the calibration plane. The practical question is therefore not only whether a planar model is geometrically imperfect, but whether the induced error is large enough to matter for the downstream task. We derive a closed-form characterization of off-plane projective bias in terms of target height and lateral camera–target geometry, and evaluate the dominant structure without fitting nonzero-height observations. Across five physical rebuilds and two cameras over a tested height range of 0–50 mm, predicted and observed degradation exhibit height-curve correlations of 0.9942–0.9951, point-level magnitude correlations of 0.9869–0.9942, and median direction cosine values of 0.9983–0.9992. The remaining geometry-unexplained vector RMSE is 3.61–4.36 mm, while independently recoverable measurement-only radial Q95 values are approximately 0.024–0.266 mm under the $\sigma h=0$ scenario, showing that geometry explains the dominant trend but not the full real-system residual. We then formulate task-conditioned adequacy under declared error scenarios using $E_{geom} + SP \leq \tau$ for planar localisation and $S3D \leq \tau$ for a 3D-aware reference. This separates task necessity from error dominance. Supporting external analysis on STAR-loc uses 12,473 repeated observations across 55 tags and eight held-out sequences and yields a Spearman magnitude correlation of 0.981 and median direction cosine of 0.998, while also exposing rare near-grazing failures. The results support a bounded, task-conditioned interpretation: the need for 3D depends on task-relevant induced error and declared system-error scenarios, not on departure from planarity alone.

Keywords— vision-guided manipulation; planar localisation; homography; projective geometry; task-conditioned adequacy; bounded error; 3D perception; calibration.

1. Introduction

Planar image-to-workspace mappings are widely used when a robot interacts with objects on a known work surface. Their appeal is practical: calibration can be lightweight, inference is inexpensive, and the resulting mapping is easy to integrate with manipulation pipelines. However, the same simplification creates a predictable weakness. Once a target is displaced from the calibration plane, the image ray associated with that target intersects the reference plane at a different lateral location, producing a systematic projective bias.

A purely geometric view makes the design choice appear simple: a model that explicitly represents 3D geometry is more faithful outside the calibration plane. Yet geometric correctness and task adequacy are not equivalent. A planar model can be wrong in a model-theoretic sense and still be adequate if its total task-relevant localisation error remains below the tolerance of the downstream manipulation. Conversely, a richer 3D-aware method can be more accurate without being necessary for task completion.

This distinction matters for low-cost or resource-constrained systems because adding 3D sensing or richer geometric estimation introduces sensing, calibration, computation, and integration costs. The engineering decision should therefore depend on the magnitude of task-relevant induced error rather than on planarity as a binary scene label.

The primary research question of this work is: under what geometric and bounded-error conditions does a planar image-to-workspace localisation model remain adequate for a specified manipulation tolerance despite off-plane model mismatch? A secondary question asks when a 3D-aware localisation method becomes the only method, among the compared bounded scenarios, that can provide the declared task guarantee.

The paper makes three contributions. First, it derives a closed-form geometry-to-error characterization of planar off-plane localisation bias and validates the dominant magnitude, direction, and spatial structure using controlled

physical experiments without fitting nonzero-height observations. Second, it formulates calibration-plane image-to-workspace localisation as a task-conditioned adequacy problem by combining an analytically characterized off-plane projective mismatch with a separately represented bounded real-system error scenario and comparing the resulting declared bound with a specified manipulation tolerance. Under the declared bounds, the framework distinguishes planar-guaranteed, 3D-only-guaranteed, and robust-error-dominance regimes without equating model accuracy with task necessity. Third, it provides multi-layer empirical support through controlled rebuilds, a residual/measurement-uncertainty audit, and an independent STAR-loc structural validation.

2. Related Work

2.1 Planar geometry, homography bias, and geometric model selection

The fact that homography-based mappings incur bias when scene points depart from the reference plane is established in projective geometry. In the SLP multimodal dataset work, Liu et al. explicitly discuss bounded cross-modal mapping bias under homography when the labelled body point is displaced from the reference plane [1]. That work is important here because it shows that bounded off-plane mapping bias itself is not a new contribution. Our focus is instead the use of such geometry within a task-conditioned model-adequacy argument that also keeps a separate real-system error scenario.

Geometric model selection is also well established. ORB-SLAM, for example, evaluates alternative geometric models during monocular initialization and chooses between them based on how well they explain image measurements and parallax [2]. Such selection asks which geometric model is appropriate for the observed scene. Our question is different: even when the planar model is not strictly correct, is its task-relevant error still small enough that a richer model is unnecessary?

2.2 Task-specific localisation requirements

Task dependence of localisation quality is likewise not new. Carlson and Kolling argue that localisation requirements should be defined relative to the downstream work task and optimize visual-fiducial placement accordingly [3]. Song et al. similarly propose task-oriented calibration for a 5-DOF hybrid machining robot by separating pose-error components that matter to the machining task from those that do not [4]. These works establish the broader task-aware principle. The present work applies that principle to a narrower localisation-model question: whether the additional fidelity of 3D-aware localisation is required once planar model mismatch and a separately declared real-system error scenario are considered together.

2.3 Non-planar floors and the closest problem-level neighbour

The closest problem-level neighbour identified in the present targeted search is Carlson's "Flatland Revisited" study [5]. It analyzes failures caused by applying a flat-floor assumption to single-image fiducial pose estimation, derives an upper bound on pose error induced by floor non-flatness, relates the resulting error to an explicit task accuracy requirement, and demonstrates a richer non-flat estimator in warehouse deployment. Consequently, neither "planar-assumption error versus task tolerance" nor the general idea of replacing an inadequate planar estimator with a richer geometric treatment is claimed here as first-ever novelty.

The remaining distinction is narrower. We explicitly separate analytically characterized planar model mismatch from a distinct bounded real-system error scenario, use the sum only as a declared conditional bound, and distinguish the regime in which only 3D is guaranteed from the logically different regime in which 3D is robustly more accurate. This distinction is then linked to controlled off-plane physical validation, a residual audit, and an external structural validation.

2.4 Bounded-error localisation and sensing/model complexity

Set-membership and interval methods have long studied localisation under unknown-but-bounded uncertainty [6]. Sensor-placement work similarly asks how much sensing is needed to achieve a required localisation accuracy; Zhao et al., for example, optimize UWB radio placement and the number of radios needed for a desired localisation accuracy [7]. More recent task-driven co-design work, such as CODEI, couples perception requirements with

sensor and algorithm selection [8]. ADAMS switches between 2D and 3D map representations according to scene structure and resource trade-offs [9]. These lines of work show that bounded guarantees, task-driven sensing complexity, and 2D/3D representation switching are not individually novel. Our applied methodological contribution is their specific integration around calibration-plane image-to-workspace localisation under off-plane projective mismatch.

3. Problem Formulation

Consider a camera with center C = (Cx, Cy, Cz) observing a target point P = (X, Y, h), where the calibrated work plane is z = 0 and h denotes the target height relative to that plane. A planar image-to-workspace mapping is calibrated only on z = 0. When it is applied to the image of the off-plane target, the corresponding camera ray is intersected with z = 0, yielding a planar estimate P′ rather than the true lateral target location.

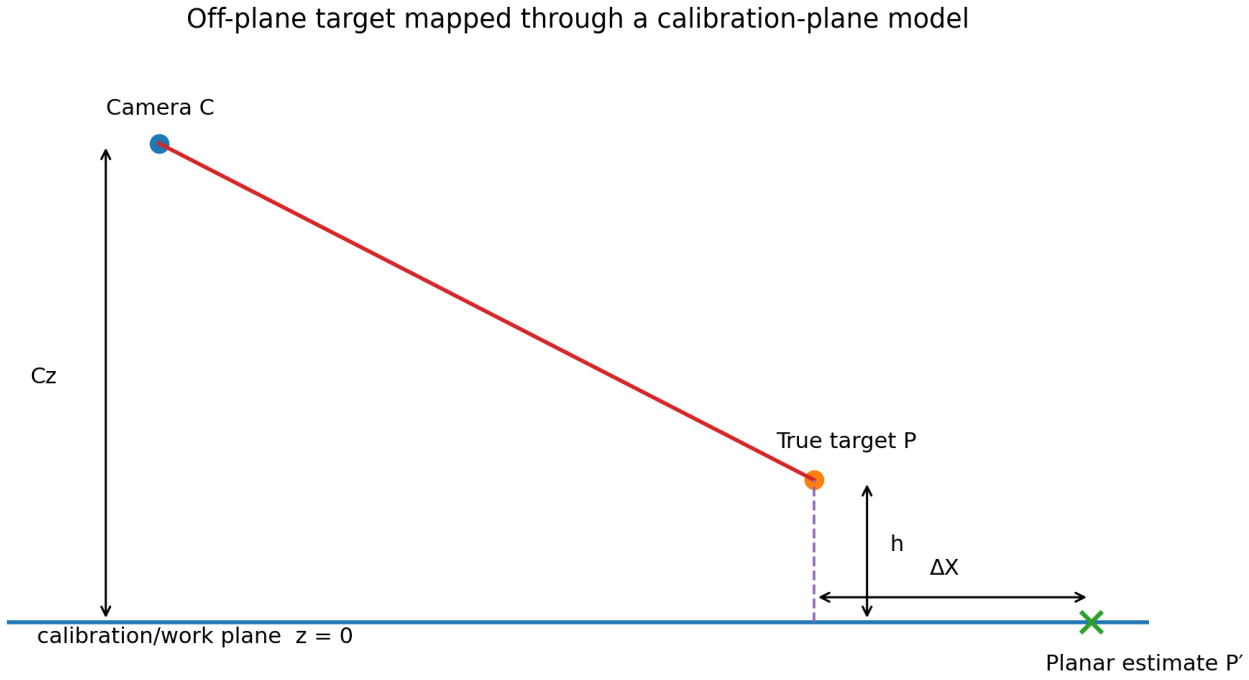


Figure 1. Geometry of off-plane bias when a calibration-plane mapping is applied to a target at height h.

Let ρ denote the lateral camera–target distance on the work plane,

$$\rho = \sqrt{[(X - Cx)^2 + (Y - Cy)^2]}. \quad (1)$$

The downstream task is represented by a work-plane XY localisation tolerance τ. We deliberately use an XY Euclidean metric because the present evidence does not establish a full SE(3) end-to-end manipulation guarantee.

A separate term SP denotes a declared planar systematic-error magnitude scenario representing non-geometric real-system error. It is not a Gaussian standard deviation, not a population-calibrated universal bound, and not inferred by fitting the unexplained residual. Similarly, S3D denotes a declared 3D-aware XY error upper-bound scenario. The scenario sets used in sensitivity analysis are SP ∈ {0, 1, 2, 3, 4, 5, 7.5, 10} mm and S3D ∈ {1, 2, 3, 5, 7.5, 10, 15} mm; these are broad sensitivity scenarios rather than universal empirical constants.

4. Off-Plane Projective Geometry

By intersecting the ray from the camera center through P with the calibration plane, the planar displacement between the true lateral position and the plane-intersection estimate is

$$\Delta X = h(X - Cx)/(Cz - h), \quad \Delta Y = h(Y - Cy)/(Cz - h). \quad (2)$$

The corresponding magnitude is

$$E_{geom} = |h|\,\rho / |C_z - h|. \quad (3)$$

Using normalized variables $\alpha = h/C_z$ and $\beta = \rho/C_z$ gives

$$E_{geom}/C_z = |\alpha|\,\beta / |1 - \alpha|. \quad (4)$$

Equation (4) is central to the interpretation. Off-plane error is not controlled by height alone; it depends jointly on relative height and lateral camera–target geometry. As h approaches Cz the ray-plane intersection becomes increasingly ill-conditioned. Therefore, fixed height thresholds such as 20 mm, 50 mm, or 5 cm cannot be interpreted as universal crossovers between planar and 3D-aware localisation.

5. Experimental Evidence

5.1 E1: in-plane baseline

E1 establishes only that the frozen planar mapping works on its calibration plane. In held-out evaluation, Homography achieved mean/RMSE/maximum XY errors of approximately 0.585/0.660/1.197 mm, while the PnP reference achieved approximately 0.587/0.661/1.293 mm. These near-equal in-plane values are treated as a baseline, not as evidence for off-plane novelty.

AUTHOR TODO: Insert E1 sample counts, exact train/held-out split, target layout, calibration details, and the Affine baseline values if retained in the final paper.

5.2 E2: controlled held-out-height validation

The principal physical validation uses five independent physical rebuilds and two cameras under a fixed-board protocol, with tested target heights spanning 0–50 mm. The projective prediction is obtained from the Z=0 geometry and is evaluated against nonzero-height degradation without fitting the held-out off-plane observations. The experiment is designed to test the structure of the geometric mechanism rather than to claim complete prediction of system accuracy.

AUTHOR TODO: Insert exact tested heights, number and coordinates of target points, frame counts, camera models, camera heights, calibration method, repeat count, exclusion criteria, and ground-truth acquisition procedure from the experiment repository.

FIGURE 2 PLACEHOLDER
Controlled E2 blind geometry validation

Recommended final panels: predicted vs observed height curves across rebuilds/cameras; point-level predicted vs observed magnitude; optional direction agreement. Use only frozen Stage 2 outputs.

Figure 2. Controlled E2 blind geometry validation. [Replace this placeholder with the frozen final figure.]

5.3 Residual and measurement-uncertainty audit

After subtracting the projective prediction, the remaining residual is not assumed to be isotropic sensor noise. The Stage 3 audit separately evaluates independently recoverable pixel, pose, and frozen-map measurement variability. Under the $\sigma h = 0$ scenario, planar measurement-only radial Q95 values are approximately 0.024–0.266 mm, substantially below the 3.61–4.36 mm geometry-unexplained vector RMSE observed in the real E2 data. The residual also exhibits directional, rebuild-specific, and height-dependent structure. The correct interpretation is therefore that the independently recoverable measurement variability is insufficient to explain the full residual; the causal source of the residual is not claimed to be identified.

An empirical PnP reference is less height-sensitive than the frozen planar mapping, but its absolute accuracy is not explained by the simple measurement model. Observed PnP XY RMSE values are approximately 9.99 mm for one camera and 5.10 mm for the other. PnP is used only as one 3D-aware reference and is not presented as the optimal 3D method.

FIGURE 3 PLACEHOLDER
What geometry does not explain

Recommended final panels: geometry-unexplained residual versus measurement-only Q95/envelope; residual structure by rebuild/height. Do not reinterpret the residual as a calibrated SP value.

Figure 3. What geometry does not explain. [Replace this placeholder with the frozen final figure.]

6. Task-Conditioned Model Adequacy

For the planar model, the declared bounded error interval is

$$\mathrm{Elower_P} = \max(0, \mathrm{Egeom} - \mathrm{SP}), \quad \mathrm{Eupper_P} = \mathrm{Egeom} + \mathrm{SP}. \quad (5)$$

A sufficient condition for planar task adequacy under the declared scenario is

$$\mathrm{GP} = [\mathrm{Egeom} + \mathrm{SP} \leq \tau]. \quad (6)$$

For the 3D-aware reference, the corresponding conditional guarantee is

$$\mathrm{G3D} = [\mathrm{S3D} \leq \tau]. \quad (7)$$

These definitions generate four guarantee classes. “Not guaranteed” does not mean the method fails; it means only that the declared bound is insufficient to certify the task tolerance.

Condition	3D-aware guaranteed	3D-aware not guaranteed
Planar guaranteed	Both guaranteed; 3D may only add accuracy	Planar-only guaranteed under declared scenarios
Planar not guaranteed	3D-only-guaranteed regime	Neither method guaranteed; failure is not implied

Figure 4. Four task-conditioned guarantee classes. The interpretation is conditional on declared SP, S3D, and τ.

Accuracy dominance is a separate statement. A sufficient robust-dominance condition for the 3D-aware method is

$$\mathrm{Egeom} > \mathrm{SP} + \mathrm{S3D}. \quad (8)$$

Equation (8) and the guarantee classes answer different questions. A 3D-aware method may be robustly more accurate while both methods remain within τ; in that case 3D improves accuracy but is not required for the task guarantee. Conversely, a 3D-only-guaranteed regime occurs only when the planar bound exceeds τ while the 3D-aware bound does not.

For a declared operating envelope with $h \leq h_{max}$ and $\rho \leq \rho_{max}$, a conservative geometry bound is

$$E_{max_geom} = h_{max}\,\rho_{max} / (C_z - h_{max}), \quad (9)$$

and planar localisation is guaranteed adequate over that envelope whenever $E_{max_geom} + SP \leq \tau$. This operating-envelope form is a practical presentation of the same bound, not a new certification algorithm or theorem.

7. Results

7.1 In-plane baseline and off-plane mechanism validation

The E1 baseline confirms that the frozen planar mapping is accurate on the calibration plane, with Homography RMSE ≈ 0.660 mm and PnP RMSE ≈ 0.661 mm. The off-plane E2 experiment then shows that Z=0 projective geometry predicts the dominant degradation structure across independent rebuilds and cameras. Height-curve correlations range from approximately 0.9942 to 0.9951, point-level magnitude correlations from 0.9869 to 0.9942, and median direction cosine from 0.9983 to 0.9992.

These values support a structural claim, not a complete-system-accuracy claim. The geometry-unexplained vector RMSE remains approximately 3.61–4.36 mm, demonstrating that a strong projective mechanism can coexist with meaningful system-specific residual error.

Evidence block	Quantity	Frozen value	Interpretation
E1	Homography mean / RMSE / max	0.585 / 0.660 / 1.197 mm	In-plane baseline only
E1	PnP mean / RMSE / max	0.587 / 0.661 / 1.293 mm	3D-aware reference baseline
E2	Physical rebuilds × cameras	5 × 2	Independent controlled rebuilds
E2	Tested height range	0–50 mm	Not a universal crossover range
E2	Height-curve correlation	0.9942–0.9951	Strong curve-level structural agreement
E2	Point-level magnitude correlation	0.9869–0.9942	Strong magnitude structure
E2	Median direction cosine	0.9983–0.9992	Strong direction agreement
E2/Stage 3	Geometry-unexplained vector RMSE	3.61–4.36 mm	Residual remains after geometry
Stage 3	Measurement-only planar radial Q95	0.024–0.266 mm	Known recoverable variability under-covers residual
Stage 3	Empirical PnP XY RMSE	9.99 / 5.10 mm	One 3D-aware reference; camera dependent

Table 1. Core quantitative evidence retained in the current manuscript draft.

7.2 Task-conditioned operating regimes

The adequacy framework converts the geometric mechanism into an engineering decision rule. Because Egeom depends on both h and ρ, the boundary between certified planar adequacy and non-certification changes with camera height, lateral workspace extent, task tolerance, and the declared SP scenario. Therefore, the framework does not support a single height-only crossover. The same target height can be benign near the camera optical axis and problematic at larger lateral distances.

The scenario sets for SP and S3D are used as sensitivity ranges. They should be reported as declared assumptions, not as probabilities or population quantiles. Under a given τ, the relevant outcome is a guarantee class rather than a success probability. A future version of the paper may add task-specific examples once τ is tied to a concrete manipulation tolerance; the current draft keeps the result conditional.

AUTHOR TODO: If a real downstream manipulation tolerance τ is available from the application, insert one or more explicit operating-regime examples. Otherwise keep all task maps scenario-based and label them as conditional sensitivity analysis.

7.3 Supporting external structural validation on STAR-loc

To test whether the structural mechanism is specific to the original low-cost camera setup, we conduct a supporting external analysis on the STAR-loc dataset [10]. The evaluation uses eight held-out setup-s3 sequences; the apriltag_s3 sequence used by the dataset authors for global calibration is excluded from evaluation. Evaluated observations use globally calibrated camera poses and measured AprilTag pixels. A reference plane is reconstructed from the known 3D positions of a low-height landmark subset; image residuals are not used to fit this plane. No off-plane observations are used to fit or rescale the projective prediction.

For each retained observation, egeom denotes the theoretical planar displacement obtained from known camera/target geometry, eobs denotes the observed planar displacement obtained by intersecting the measured image ray with the externally reconstructed reference plane, and r = eobs − egeom is the residual. The main metrics are magnitude-ordering correlation, direction cosine, the residual-to-observed magnitude ratio, and ray-plane conditioning.

$$\gamma_{obs} = |n^T \hat{d}_{obs}|, \quad (10)$$

where smaller γ_{obs} indicates a more grazing measured ray and a more ill-conditioned ray-plane intersection.

The retained data comprise 12,473 repeated observations across 55 tags and eight sequences. Spearman correlation between predicted and observed displacement magnitudes is 0.981, and the median direction cosine is 0.998. At the observation level, $P(R < D) = 88.65\%$, $P(R < 0.5D) = 75.54\%$, and the median R/D is 0.183, where D is the observed displacement magnitude and R is the residual magnitude.

Pooled squared-error metrics are dominated by a very small catastrophic tail: the top 13 observations (approximately 0.1%) contribute about 99.79% of residual SSE. The largest failures are associated with near-grazing measured rays and extremely ill-conditioned ray-plane intersections. Thus, STAR-loc supports cross-system generalisation of the structural projective mechanism while simultaneously exposing a failure regime in which small ray perturbations are dramatically amplified.

The external dataset does not provide a clean independent validation of absolute error scale. The reconstructed low-landmark reference plane differs non-negligibly from an alternative Vicon-horizontal definition; therefore, STAR-loc is used only as structural supporting evidence, not as an absolute-accuracy replication, an estimator of SP, or a second task-boundary experiment.

FIGURE 5 PLACEHOLDER
STAR-loc external structural validation

Recommended final panels: (A) predicted vs observed magnitude on log-log axes; (B) direction cosine versus geometric signal magnitude; (C) residual magnitude versus γobs. Main text should show structure, signal identifiability, and failure regime; full tail/plane/macro audits belong in Supplementary Material.

Figure 5. STAR-loc external structural validation. [Replace this placeholder with the frozen final figure.]

STAR-loc quantity	Value
Sequences	8
Tags	55
Retained observations	12,473 repeated observations
Spearman(Egeom, Eobs)	0.981
Median direction cosine	0.998

STAR-loc quantity	Value
P(R < D)	88.65%
P(R < 0.5D)	75.54%
Median R/D	0.183
Residual SSE from top ≈0.1%	99.79%
Absolute-scale verdict	Not cleanly identifiable

Table 2. Compact STAR-loc external-validation summary retained for the main manuscript.

8. Discussion

8.1 Model correctness is not model adequacy

The central implication is semantic as much as numerical. A planar model ceases to be geometrically exact as soon as an observed target departs from the calibration plane, but this does not imply that it ceases to be useful. The engineering question is whether the induced error, together with a declared allowance for non-geometric system error, violates the task tolerance. This reframes the planar-versus-3D decision from a binary scene-classification problem into a task-conditioned operating-regime problem.

8.2 Height alone is not a sufficient switching variable

Both the closed-form model and the external analysis argue against height-only switching rules. The geometric term scales with relative height and lateral camera–target geometry, while external failures also reveal a conditioning dependence. Consequently, a fixed statement such as “3D is required above 50 mm” is not supported. A declared operating boundary must specify the workspace geometry, camera placement, task tolerance, and error scenario.

8.3 Why the residual must remain separate

The Stage 3 audit is important because it prevents a common interpretive shortcut. The strong geometry correlations might otherwise encourage the claim that the real-system error is essentially solved by projective analysis. That claim is not supported: the residual RMSE remains several millimetres and is much larger than the independently recoverable measurement-only variability under the tested audit assumptions. Keeping this residual separate from Egeom prevents circular calibration of SP and makes the final guarantee explicitly conditional.

8.4 3D accuracy versus 3D necessity

A richer 3D-aware method can be preferred for many reasons, including lower error, robustness to wider scene classes, or easier extension to full 3D tasks. The narrower claim here is that “more accurate” and “required for a task guarantee” are not equivalent. If both GP and G3D hold, 3D may offer extra accuracy without being necessary for the declared task tolerance. If GP is false but G3D is true, only then does the declared scenario enter a 3D-only-guaranteed regime. This separation is intentionally conservative and does not imply that the planar method fails whenever GP is false.

8.5 Relation to closest work

The targeted closest-work audit suggests that neighbouring literature covers different subsets of the present chain. SLP provides bounded homography-related mapping bias [1]; ORB-SLAM provides geometric model selection [2]; Selective Fiducials and task-oriented calibration establish task-dependent error requirements [3], [4]; Flatland Revisited directly relates a planar-assumption error bound to a task accuracy requirement and develops a richer estimator [5]; bounded-error localisation provides set-based guarantees [6]; and sensor/algorithm co-design addresses task-driven sensing complexity [7]–[9]. The present paper should therefore not use first-ever language. Its

contribution is the specific integration and separation of these elements for calibration-plane image-to-workspace localisation, together with the particular empirical evidence chain presented here.

9. Limitations

- The controlled E2 validation uses five physical rebuilds. This is sufficient for the present mechanism-validation claim but not for population-level distribution estimation.
- SP and S3D are declared sensitivity scenarios, not calibrated universal bounds or probabilities. All guarantee statements are conditional on these declarations.
- PnP is one 3D-aware reference and should not be interpreted as the optimal 3D localisation method.
- The primary metric is board/work-plane XY Euclidean localisation error. The results do not establish full SE(3) pose accuracy, grasp success, or end-to-end manipulation success.
- The causal source of the geometry-unexplained residual is not identified. The Stage 3 result only establishes that the independently recoverable measurement-only variability is insufficient to explain it.
- STAR-loc supports structural mechanism generalisation but not clean absolute-scale replication because the operational reference plane is externally reconstructed and its definition is not unique.
- The closest-work audit is targeted rather than systematic. The manuscript should use “we formulate”, “we establish”, and “we show” rather than first-ever or exhaustive novelty claims.
- No universal 20 mm, 50 mm, or 5 cm crossover is claimed, and “not guaranteed” is not interpreted as failure.

10. Conclusion

This paper studies when a calibration-plane image-to-workspace model remains adequate after its planar assumption is violated. The off-plane bias admits a simple projective characterization whose dominant magnitude, direction, and spatial structure are strongly supported by controlled physical experiments across rebuilds and cameras. At the same time, a residual audit shows that geometry does not explain the complete real-system error. This motivates a task-conditioned framework in which the analytically characterized mismatch is combined with a separately declared real-system error scenario and compared with an explicit task tolerance. The resulting guarantee classes distinguish planar adequacy, 3D-only guarantee regimes, and robust accuracy dominance without conflating them. External STAR-loc analysis supports the structural projective mechanism across a different system while also identifying near-grazing ray-plane intersections as a severe failure regime. The main engineering conclusion is therefore bounded: the need for 3D localisation is determined by task-relevant induced error, declared real-system error scenarios, workspace geometry, and conditioning—not by departure from planarity alone.

Data and Code Availability

AUTHOR TODO: Insert repository URL(s), archived commit/tag, dataset provenance statement, and any code/data release restrictions. Cite STAR-loc according to its dataset paper [10].

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Appendix A. Author Completion Checklist (remove before submission)

- Authors, affiliations, corresponding author, acknowledgements/funding.
- Exact hardware names and setup geometry.
- E1/E2 sample counts, exact height list, target positions, frame counts, exclusions, ground-truth method.
- Final frozen Figure 2 (E2), Figure 3 (residual audit), Figure 5 (STAR-loc).
- Concrete downstream tolerance τ, only if justified by the intended manipulation task; otherwise keep the framework scenario-based.
- Any additional target-journal literature required by the supervisor and final bibliography style verification.
- Repository/code availability statement and frozen commit/tag.
- Final reviewer-style traceability audit: every quantitative claim should map to a frozen dataset/report/figure.
- Final target-journal formatting, word count, figure count, and reference style.